
Does the Decoding Algorithm Have a Voice? Measuring the Sampler’s Stylistic Signature Against the Prompt’s

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Abstract

1 When a language model writes a poem, the decoding algorithm (greedy, tempera-
2 ture, top- k , nucleus, or min- p sampling) is usually chosen by a platform default
3 rather than by the person prompting. We ask whether this silent choice has a
4 measurable stylistic voice of its own, independent of the model and the prompt.
5 Holding the model, prompt set, system prompt, token budget, and seed schedule
6 fixed, we vary only the sampler across ten configurations in five families, generat-
7 ing 2,160 poems in a fully crossed sampler $\times$ prompt $\times$ model design over three
8 open-weight models. A classifier recovers the sampler from 29 style features at
9 0.139 accuracy under leave-one-prompt-out cross-validation (chance 0.100, permuta-
10 tion $p = 0.005$), and the effect survives dropping every length feature. But sizing
11 the effect matters more than detecting it: under an identical protocol, the sampler’s
12 classification skill is ~ 0.04 against 0.44 for the model and 0.77 for the prompt, and
13 an independent variance decomposition agrees (median shares 0.021 vs. 0.189 for
14 the prompt). The trace is also concentrated: high temperature and greedy decoding
15 account for nearly all of it, while nucleus, top- k , and min- p at conventional settings,
16 the knobs platforms actually ship, are mutually indistinguishable by this battery.
17 The decoding algorithm is a real but minor co-author: a systematic nudge roughly
18 one-twentieth the size of the human’s prompt, on a process whose largest variance
19 component is sampling noise itself.

20 1 Introduction

21 Every generation from a language model passes through a decoding algorithm. Greedy decoding
22 takes the argmax; temperature rescales the logits; top- k [Fan et al., 2018], nucleus [Holtzman et al.,
23 2020], and min- p [Nguyen et al., 2024] sampling truncate the distribution in different geometries
24 before sampling from what remains. In deployed systems this choice is almost never made by the
25 person writing the prompt. It is a platform default, set once and applied silently to everything the
26 system produces.

27 This raises a question about creative agency that is empirical rather than philosophical: does the
28 decoding algorithm have a stylistic voice of its own? If the sampler that produced a text can be
29 recovered from the text’s style alone, independent of the model and the prompt, then a decision the
30 author never made is measurably co-authoring the output. If it cannot, the sampler is stylistically
31 transparent and authorship talk can safely ignore it.

32 Detection alone, however, is a weak finding: with enough data, almost any manipulation is detectable.
33 The question that matters for agency is one of *proportion*: how does the sampler’s contribution
34 compare to that of the prompt, i.e. to the part the human actually wrote? We therefore run a fully
35 crossed sampler $\times$ prompt $\times$ model experiment and measure all three factors in a common currency.

Our findings cut both ways for the agency theme. The sampler does leave a statistically robust stylistic trace, detectable on prompts the classifier has never seen. But it is one to two orders of magnitude smaller than the prompt’s trace, it is concentrated almost entirely in two extreme configurations (greedy and temperature 1.3), and the majority of stylistic variance belongs to neither the human nor the configuration: it is sampling noise. Nucleus, top- k , and min- p at conventional settings, the defaults most platforms actually ship, are stylistically interchangeable by our feature battery. Agency over a generated poem’s style turns out to be diffuse rather than contested: dominated by the prompt where it is systematic at all, and dominated by chance where it is not.

2 Experimental design

One independent variable. We hold fixed the system prompt (“You are a poet. Respond with the poem only...”), the token budget (`max_tokens = 320`), the prompt set, and the seed schedule, and vary only the sampler. Ten configurations span five families: greedy ($T=0$); pure ancestral sampling ($T=1.0$) plus temperature 0.7 and 1.3; top- k at $k \in \{20, 50\}$; nucleus at $p \in \{0.90, 0.95\}$; and min- p at $\{0.05, 0.10\}$. Every request pins *all four* shaping parameters explicitly, with unused knobs at their neutral no-op values ($T=1.0$, top- $p=1.0$, top- $k=0$, min- $p=0.0$). Sending only top- $p=0.9$ and letting the provider default the rest would compare nucleus-plus-whatever-the-default-top- k -is against min- p -plus-the-same; pinning every knob makes each configuration a clean single-parameter contrast, and $T=1.0$ doubles as the shared baseline every truncation method reduces to at its neutral setting.

Three crossed factors. Twelve poem prompts span a constraint gradient, from four open (“Write a poem about a city at dawn”) through four moderately constrained to four tightly constrained (“Write a sonnet about a broken clock”), because a fingerprint that appears only under loose prompts is a weaker claim than one that survives when the prompt is doing most of the work. Three open-weight models from three families (gpt-oss-20b, nemotron-3-ultra-nvfp4, deepseek-v4-flash) are queried through a single provider (Fireworks), which exposes temperature, top- p , top- k , and min- p on one OpenAI-compatible route; mixing providers per sampler would confound the fingerprint with serving differences. The model factor is what licenses the word “sampler” in the conclusion: a fingerprint on one checkpoint could be a property of that model’s logit geometry rather than of decoding, and an effect only counts for us if it points the same direction on all three.

With 6 repeats per cell the design is $10 \times 12 \times 3 \times 6 = 2,160$ generations, fully crossed. The random seed for a cell is a function of (`prompt_id`, `rep`) only, so every sampler on every model sees an identical seed schedule, so a difference between samplers cannot be an artifact of different random streams.

Features. Each poem is reduced to 29 style features, all rates or distributional statistics, never raw counts, so a sampler that simply writes more does not score higher on everything. They cover lexical diversity (TTR, length-corrected MATTR-50 [Covington and McFall, 2010], hapax rate, unigram entropy, function-word rate, mean Zipf word frequency [Speer et al., 2018]), line structure (line/stanza statistics, enjambment, punctuation rates), and imagery via a sensory-modality lexicon.

Tests, hardest last. (1) *Univariate*: Kruskal–Wallis per feature with ϵ^2 effect sizes, Benjamini–Hochberg FDR across the battery. (2) *Fingerprint*: multinomial logistic regression recovering the sampler from the features, reported under **leave-one-prompt-out** (LOPO) cross-validation: train on eleven prompts, test on one never seen, so the classifier cannot win by memorizing topic–sampler associations. (3) *Null*: labels permuted *within prompt* (preserving prompt structure and cell balance) and the full LOPO pipeline rerun 200 times, giving an empirical null and a permutation p -value. (4) *Ablation*: the same test with all seven length features dropped, since a sampler that merely writes longer poems is a weak claim about voice. The analysis pipeline was validated on synthetic data with planted effects, including a required *failure* on data with no planted effect. This is the check that would have caught a false positive.

Sizing the effect two ways. First, *common-currency classification*: the same features, classifier, and CV protocol pointed at three targets, each holding out levels of a nuisance factor (predict sampler holding out a prompt; predict model holding out a prompt; predict prompt holding out a sampler). We report *skill*, defined as accuracy above chance as a share of available headroom, so tasks with 10, 12, and 3 classes land on one scale. Second, a balanced crossed random-effects *variance decomposition*

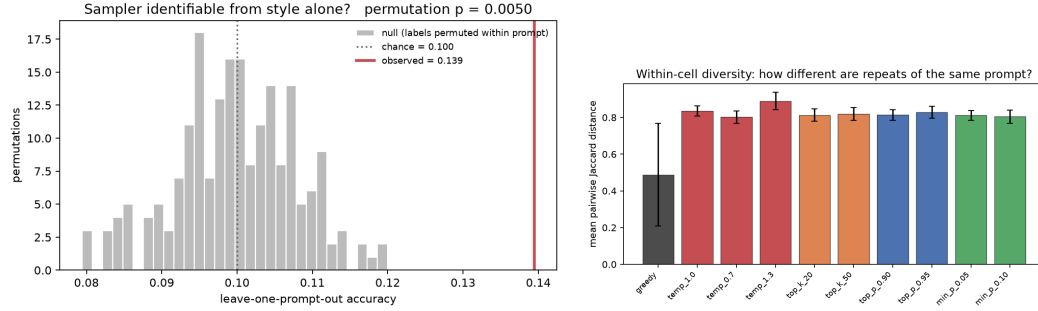


Figure 1: **Left:** observed LPO accuracy (run B) against 200 within-prompt permutation nulls; the sampler fingerprint is real ($p = 0.005$). **Right:** within-cell diversity (mean pairwise Jaccard distance across the six repeats of one prompt under one sampler; error bars are standard deviations across cells). Greedy collapses toward self-repetition (0.49) while every stochastic sampler sits at 0.80–0.89, a designed sanity check that the pipeline is wired correctly, and a style signal in its own right.

per feature, sharing none of the classifier’s machinery; interactions are pooled into the residual, which makes every factor share a lower bound, conservative in the direction that matters, since it cannot inflate the sampler’s apparent contribution.

Three analysis runs. Run A uses the full corpus of 2,085 poems (75 of 2,160 generations failed; see Limitations). Run B drops two prompts to obtain a fully balanced $10 \times 10 \times 3 \times 6$ design of 1,800 poems, which is what permits the variance decomposition; run C keeps only the 1,670 generations with `finish_reason = "stop"` as a truncation control. Unless noted, figures show run B; we report A and C as brackets around it.

3 Results

3.1 The sampler is detectable...

The LPO classifier recovers the sampler at accuracy 0.139 in run B against a chance rate of 0.100 (0.145 in run A, 0.120 in run C), with permutation $p = 0.005$ in all three runs: the observed accuracy exceeds all 200 within-prompt permutation nulls (Figure 1, left). The effect survives dropping every length feature (run B: 0.144, $p = 0.0099$), so the sampler is changing how poems *sound*, not merely how much text comes out. At the univariate level, 25 of 29 features differ significantly across samplers at $FDR < 0.05$ in run B, led by lexical diversity (MATTR-50, $\varepsilon^2 = 0.054$), mean Zipf frequency, and comma rate. High temperature drags vocabulary toward rarer, longer words with fewer function words and more enjambment; greedy decoding does the opposite.

3.2 ...but it is small, and the prompt dominates

Detection is not the finding; proportion is. Under the identical protocol, predicting the *prompt* from style (generalizing across held-out samplers) reaches skill 0.77, predicting the *model* (across held-out prompts) reaches 0.44, and predicting the *sampler* reaches 0.04 (Figure 2, left). The sampler-to-prompt skill ratio is 0.057 in run B and lies in $[0.027, 0.066]$ across all three runs: the human-authored prompt is stylistically legible at fifteen to forty times the strength of the decoding configuration.

The variance decomposition, which shares none of the classifier’s machinery, agrees. Median shares across features: residual 0.712, prompt 0.189, model 0.033, sampler 0.021 (Figure 2, right). The sampler is never the dominant factor for any single feature; its largest share anywhere is 0.080, on mean Zipf frequency. Convergence of two independent measures that fail in different ways is, we think, the strongest evidence in this paper. Note also what the residual says: at 0.712, most of what distinguishes two poems from the very same cell is neither the human nor the configuration: it is sampling noise.

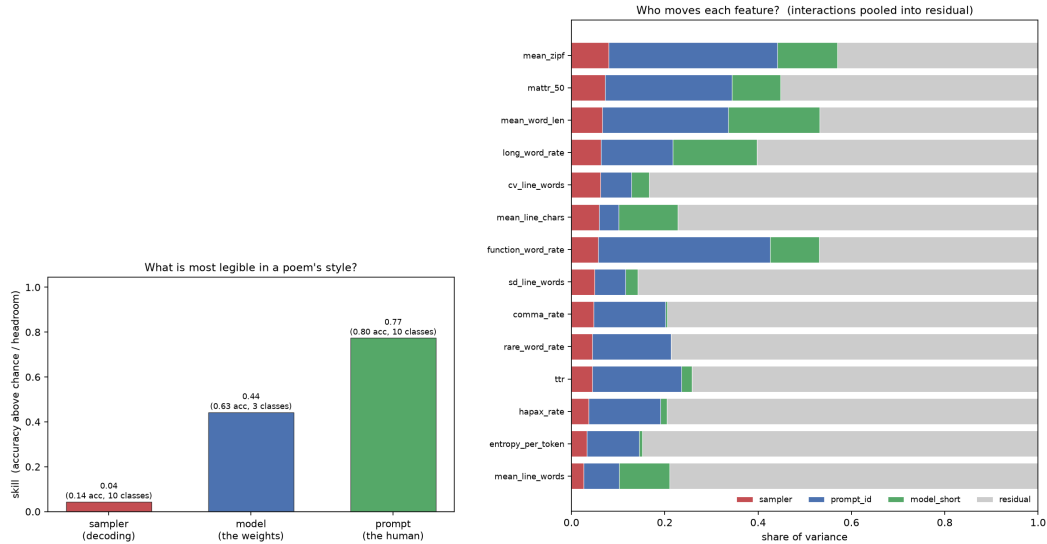


Figure 2: **Left:** the headline comparison. The same 29 features, classifier, and held-out-nuisance CV protocol pointed at three targets, on one skill scale (accuracy above chance / headroom). The prompt dominates; the sampler is real but an order of magnitude smaller. **Right:** per-feature variance shares from a balanced crossed random-effects decomposition (run B), interactions pooled into the residual so every colored share is a lower bound. The sampler (red) never leads any feature.

3.3 The trace lives in two samplers, not ten

The fingerprint is not spread evenly. In the run-B confusion matrix (Figure 3, left), temperature 1.3 is recovered at 0.51 and greedy at 0.27, while the other eight samplers sit between 0.04 and 0.13 against a 0.10 chance line. Pairwise separability agrees (Figure 3, right): every pair *not* involving greedy or temp 1.3 sits at 0.46–0.55 binary LOPO accuracy, which is chance. Nucleus at 0.90 vs. min- p at 0.05, top- k 20 vs. top- k 50, top- p 0.95 vs. min- p 0.10: all stylistically indistinguishable by this battery. We consider this the sharpest claim in the paper, because these are precisely the settings platforms actually ship: the practically deployed region of decoding-parameter space has no detectable voice of its own, and only the extremes, determinism and high temperature, are legible.

Directional consistency across the three models is 0.30–0.37: only about a third of significant (feature, sampler) effects point the same way on all three checkpoints. The effects that do (temp 1.3 raising lexical diversity and word length while depressing frequency, commas, and function words; greedy doing the reverse) are the ones underwriting the confusion diagonals, and are the only individual effects we would defend as properties of *decoding* rather than of a particular model.

4 Limitations

Attrition. 2,160 generations were attempted and 2,085 analyzed; all 75 failures occurred on gpt-oss-20b, whose reasoning-style responses sometimes place output under a different response key than our parser expected. **Truncation is confounded with the sampler.** 415 of 2,085 poems hit the 320-token cap, and the rate tracks temperature (0.10 for greedy up to 0.34 for temp 1.3; by model, 0.03–0.36). A poem cut off mid-line has a ragged final line and no terminal punctuation, which inflates enjambment and line-length variability, two headline temp 1.3 effects, and the no-length ablation does not control for this, because it drops length *features*, not truncated *poems*. Run C (stop-only) addresses it, but since truncation is caused by the treatment, filtering on it selects on a post-treatment variable; we therefore report A and C as brackets (the sampler effect lies somewhere in 0.120–0.145 LOPO accuracy and is significant under both readings) rather than presenting C as the clean answer. The real fix is generating to a stop condition rather than a token cap, which is future work. **Scope.** Results cover one genre (poetry in English), one prompt style, three mid-size open-weight checkpoints, and one 29-feature battery; a richer representation (e.g., learned embeddings)

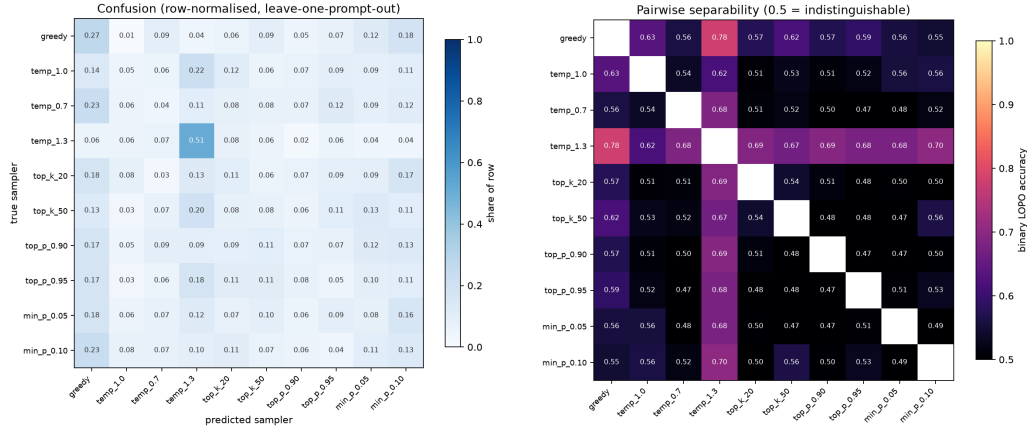


Figure 3: **Left:** row-normalized LOPO confusion (run B). Only temp_1.3 (0.51) and greedy (0.27) exceed the 0.10 chance line meaningfully. **Right:** pairwise binary separability. The bright row/column belong to temp_1.3; the interior block of nucleus, top- k , and min- p at conventional settings is at chance (0.46–0.55).

might recover more sampler signal, so our magnitudes are lower bounds relative to this battery, not ceilings. Run B drops two of the twelve prompts (the haiku sequence and the fourteen-line address to a dead language) to balance the design; run A’s classification results use all twelve. Imagery features rest on a bundled sensory-modality lexicon and corpus frequency norms, not on external human concreteness ratings.

5 Conclusion: agency is diffuse

A decoding parameter chosen by a platform default, not by the person writing the prompt, measurably shapes what the model writes: detectably, robustly, across three model families, on prompts the classifier never saw. And it does so at roughly one-twentieth the magnitude of the prompt itself, concentrated almost entirely in configurations (greedy, temperature 1.3) that deployed systems rarely ship, while the defaults they do ship are stylistically interchangeable. For questions of creative agency the result cuts both ways, and both directions are worth stating plainly: the human’s prompt dominates everything the configuration contributes, which is reassuring for authorship claims; but the largest variance component of all is the residual: two poems from the same prompt, model, and sampler differ mostly by chance. The sampler is a small systematic nudge on a very noisy process. Agency over a generated poem’s style is not so much contested between human and machine as it is diffuse: mostly the human’s where it is anyone’s, and mostly no one’s at all.

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